

Experiment 9: Run a Container from Docker Hub

Aim:

To **run a Docker container** using an image pulled from **Docker Hub**, test its execution, and understand container lifecycle management.

Requirements:

- A system with **Docker installed**
- Internet connection
- Docker Hub account (optional, for pulling public images)

Procedure:

Step 1: Verify Docker Installation

```
docker --version
```

If Docker is not installed:

```
sudo apt update  
sudo apt install docker.io -y  
sudo systemctl start docker  
sudo systemctl enable docker
```

To run Docker without sudo:

```
sudo usermod -aG docker $USER  
newgrp docker
```

Step 2: Search for an Image on Docker Hub

Example: Search for the **nginx** image

```
docker search nginx
```

Step 3: Pull an Image from Docker Hub

```
docker pull nginx
```

You can also try:

```
docker pull hello-world
```

Step 4: Run a Container

Run **hello-world** container:

```
docker run hello-world
```

This verifies Docker setup with a test container.

Run **nginx** container (web server):

```
docker run -d -p 8080:80 nginx
```

- -d runs the container in detached mode
- -p maps port 8080 (host) to 80 (container)

Open browser and go to:

👉 <http://localhost:8080>

You will see the **Nginx welcome page**.

Step 5: List Running Containers

```
docker ps
```

List all containers (even stopped):

```
docker ps -a
```

Step 6: Stop a Running Container

Get container ID:

```
docker ps
```

Then stop:

```
docker stop <container_id>
```

Step 7: Remove a Container and Image

```
docker rm <container_id>
```

```
docker rmi nginx
```

Result:

Docker image successfully pulled from Docker Hub and executed as a container.

Sample Viva Questions:

1. What is Docker and how is it different from a VM?
2. What is Docker Hub?
3. What is the difference between Docker image and Docker container?
4. How can you expose a container to external traffic?
5. What happens when you run `docker run hello-world`?